

$$1 \text{ Actuali } \int \frac{5t-4}{t+1} dt$$

$$v = t+1$$

$$t = v-1$$

$$5(v-1)-4$$

$$5v-5-4=5v-9$$

$$dv = dt$$

$$\begin{aligned} \int \frac{5v-9}{v} dv &= \int \frac{5v}{v} - \frac{9}{v} dv \\ &= \int 5 - \frac{9}{v} dv = 5 \int dv - 9 \int \frac{1}{v} dv \\ &= 5 \int dv - 9 \int \frac{1}{v} dv \\ &= 5v - 9 \ln|v| + C \\ &= 5(t+1) - 9 \ln|t+1| + C \\ &= 5t + 5 - 9 \ln|t+1| + C \end{aligned}$$

Practice:

$$\int \frac{t-6}{t+1} dt$$

$$u = t+1$$

$$t = u-1$$

$$t(u-1)-6$$

$$tu - t - 6 = tu - 7$$

$$\frac{du}{dt} = 1$$

$$du = dt$$

$$dv = dt$$

$$\int \frac{tu-7}{u} du = \int 7 - \frac{7}{u} du$$

$$\frac{tu-7}{u} = \frac{tu}{u} - \frac{7}{u} = t + \frac{7}{u} = 7$$

$$= 7u - 7 \ln|u| + C$$

$$= 7(t+1) - 7 \ln|t+1| + C$$

$$= 7t + 7 - 7 \ln|t+1| + C$$

$$= 7t + 7 - 7 \ln|t+1| + C$$

2. Actual: $\int_0^1 \frac{x-6}{x^2-6x+8} dx$

Completing the square

$$\begin{aligned} x^2 - 6x + 8 &= 0 \\ x^2 - 6x &= -8 \\ x^2 - 6x + 9 &= -8 + 9 \\ (x-3)^2 &= 1 \end{aligned}$$

$$\begin{aligned} (x-3)(x-3) \\ x^2 - 3x - 3x + 9 \\ x^2 - 6x + 9 \end{aligned}$$

$$\int_0^1 \frac{x-6}{(x-3)^2-1} dx$$

$$\begin{aligned} u &= x-3 & du &= 1 \\ u-3 &= x-3-3 & dx & \\ x-6 &= u-3 & du &= dx \end{aligned}$$

$$\int_0^1 \frac{u-3}{u^2-1} du = \int_0^1 \frac{u-3}{(u-1)(u+1)} du = \frac{A}{u+1} + \frac{B}{u-1} = \frac{A(u-1)}{(u+1)(u-1)} + \frac{B(u+1)}{(u-1)(u+1)}$$

$$\begin{aligned} (u+1)(u-1) \\ u^2 - u + u - 1 \\ u^2 - 1 \end{aligned}$$

$$u-3 = A(u-1) + B(u+1)$$

when $u=1$: $1-3 = A(1-1) + B(1+1)$
 $-2 = A(0) + B(2) \rightarrow -2 = 2B \rightarrow B = -1$

when $u=-1$: $-1-3 = A(-1-1) + B(-1+1)$
 $-4 = A(-2) + B(0) \rightarrow -4 = -2A \rightarrow A = 2$

$$\int_0^1 \frac{2}{u+1} + \frac{-1}{u-1} = 2 \int_0^1 \frac{1}{u+1} du - \int_0^1 \frac{1}{u-1} du = 2[\ln|u+1|]_0^1 - [\ln|u-1|]_0^1$$

$$\begin{aligned} u &= x-3 \rightarrow u+1 = x-3+1 = x-2 \\ u-1 &= x-3-1 = x-4 \end{aligned}$$

$$2[\ln|x-2|]_0^1 - [\ln|x-4|]_0^1$$

$$(2[\ln|1-2|] - 2[\ln|0-2|]) - (\ln|1-4| - \ln|0-4|)$$

$$(2[\ln|-1|] - 2[\ln|-2|]) - (\ln|-3| - \ln|-4|)$$

$$(2\ln|\frac{1}{2}| - 2\ln|\frac{2}{1}|) - (\ln|\frac{3}{1}| - \ln|\frac{4}{1}|) = 2\ln(\frac{1}{2}) - \ln(\frac{3}{4})$$

$$3 \text{ Actual} - \int_0^3 \frac{x^2+x+1}{(x+1)^2(x+2)} dx$$

$$\frac{x^2+x+1}{(x+1)^2(x+2)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x+2}$$

Mul both sides by $(x+1)^2(x+2)$

$$x^2+x+1 = A(x+1)(x+2) + B(x+2) + C(x+1)^2$$

$$x=-1: 1-1+1 = \cancel{A(0)(1)} + B(1) + \cancel{C(0)^2}$$

$$1 = B$$

$$x=-2: 4-2+1 = \cancel{A(-1)(0)} + \cancel{B(0)} + C(-1)^2$$

$$3 = C(1) \rightarrow 3 = C$$

$$x=0: 1 = A(1)(2) + B(2) + C(1)^2$$

$$1 = 2A + 2B + C$$

$$1 = 2A + 2 + 3$$

$$1 = 2A + 5 \rightarrow -4 = 2A \rightarrow -2 = A$$

$$\int_0^3 \frac{-2}{x+1} + \frac{1}{(x+1)^2} + \frac{3}{x+2} dx$$

$$= \left[2 \int \frac{1}{x+1} dx + \int \frac{1}{(x+1)^2} dx + 3 \int \frac{1}{x+2} dx \right]_0^3$$

$$= \left[-2 \ln|x+1| - \frac{1}{x+1} + 3 \ln|x+2| \right]_0^3$$

$$3+1=4$$

$$0+1=1$$

$$3+1=4$$

$$0+1=1$$

$$3+2=5$$

$$0+2=2$$

$$= \left[-2 \ln|4| - \frac{1}{4} + 3 \ln|5| \right] - \left[-2 \ln|1| - 1 + 3 \ln|2| \right]$$

$$= (-2 \ln|4| - \frac{1}{4} + 3 \ln|5|) - (0 - 1 + 3 \ln|2|)$$

$$= (-2 \ln|4| - \frac{1}{4} + 3 \ln|5|) - (-1 + 3 \ln|2|)$$

$$\int_0^2 \frac{x^2 + x + 1}{(x+1)^2 (x+2)} dx$$

$$(x^2 + x + 1) = (x+1)^2 - x$$

$$\int_0^2 \frac{(x+1)^2 - x}{(x+1)^2 (x+2)} dx = \int_0^2 \frac{\cancel{(x+1)^2}}{\cancel{(x+1)^2} (x+2)} - \frac{x}{(x+1)^2 (x+2)} dx$$

$$= \int_0^2 \frac{1}{x+2} - \frac{x}{(x+1)^2 (x+2)} dx = \int_0^2 \frac{1}{x+2} dx - \int_0^2 \frac{x}{(x+1)^2 (x+2)} dx$$

$$= \left[\ln|x+2| \right]_0^2 - \int_0^2 \frac{x}{(x+1)^2 (x+2)} dx$$

$$\frac{A}{x+1} + \frac{A_1}{(x+1)^2} + \frac{C}{x+2}$$

$$\begin{aligned} (x+1)(x+1) &= x^2 + 2x + 1 \\ x^2 + 2x + 1 - x &= x^2 + x + 1 \end{aligned}$$

$$4 \int \frac{9x^2 + 5x + 9}{(x^2 + 1)^2} dx$$

$(x^2 + 1)^2$
 $= (ax^2 + bx + c)^k$
 where $b=0, a=1, c=1$
 $k=2$

$$\frac{Ax+B}{x^2+1} + \frac{Cx+D}{(x^2+1)^2}$$

$$\frac{0x+9}{x^2+1} + \frac{5x+0}{(x^2+1)^2}$$

$$= \frac{9}{x^2+1} + \frac{5x}{(x^2+1)^2}$$

$$9x^2 + 5x + 9 = (Ax+B)(x^2+1) + (Cx+D)$$

$$0x^3 + 9x^2 + 5x + 9 = Ax^3 + Ax + Bx^2 + B + Cx + D$$

$$0x^3 + 9x^2 + 5x + 9 = Ax^3 + Bx^2 + x(A+C) + B+D$$

$$A=0 \quad B=9 \quad 5=A+C \quad B+D=9$$

$$0+C=5 \quad 9+D=9$$

$$C=5 \quad 9+0=9$$

$$D=0$$

$$= 9 \int \frac{1}{x^2+1} dx + 5 \int \frac{1}{(x^2+1)^2} x dx$$

$$= 9 \int \frac{1}{x^2+1} dx + 5 \int \frac{1}{(x^2+1)^2} x dx$$

$u = x^2+1$ $du = 2x dx$

$$5 \int \frac{1}{u^2} \cdot \frac{1}{2} du = 5 \left[\frac{1}{2} \int \frac{1}{u^2} du \right]$$

$x dx = \frac{1}{2} du$

$$5 \left(-\frac{1}{2u} \right) = -\frac{5}{2(x^2+1)}$$

$$\int u^{-2} = 5 \left[\frac{1}{2} \left(-\frac{1}{u} \right) \right]$$

$$= 5 \left(-\frac{1}{2u} \right)$$

$$9 \int \frac{1}{x^2+1} dx$$

$$= 9 \arctan x$$

Trig
 identity from my Calc 2 notes

$$= 9 \arctan x - \frac{5}{2(x^2+1)} + C$$

$$\int \frac{4x^2 + 7x + 4}{(x^2 + 1)^2} dx$$

$= (ax^2 + bx + c)^k$
 where $b=0, a=1, c=1$
 $k=2$

$$\frac{Ax+B}{x^2+1} + \frac{Cx+D}{(x^2+1)^2} \left\} 4x^2 + 7x + 4 = (Ax+B)(x^2+1) + Cx+D$$

$$4x^2 + 7x + 4 = Ax^3 + \underline{Ax} + Bx^2 + B + \underline{Cx} + D$$

$$0x^3 + 4x^2 + 7x + 4 = Ax^3 + Bx^2 + x(A+C) + B+D$$

$$A=0$$

$$B=4$$

$$A+C=7$$

$$B+D=4$$

$$0+C=7$$

$$4+D=4$$

$$C=7$$

$$D=0$$

$$\frac{0x+4}{x^2+1} + \frac{7x}{(x^2+1)^2} = \int \frac{4}{x^2+1} + \frac{7x}{(x^2+1)^2} dx$$

$$= 4 \int \frac{1}{x^2+1} dx + 7 \int \frac{1}{(x^2+1)^2} x dx$$

$$u = x^2+1$$

$$du = 2x dx$$

$$x dx = \frac{1}{2} du$$

$$7 \int \frac{1}{u^2} \cdot \frac{1}{2} du$$

$$7 \left(-\frac{1}{2u} \right) = -\frac{7}{2(x^2+1)}$$

$$4 \int \frac{1}{x^2+1} dx$$

Try
 identity from notes

$$= 4 \arctan x$$

$$= 4 \arctan x - \frac{7}{2(x^2+1)} + C$$

$$5 \quad \text{Actual} - \int \frac{x^4 + 15x^2 + 1}{x^5 + 25x^3 + 5x} dx$$

$$u = x^5 + 25x^3 + 5x$$

$$du = 5x^4 + 75x^2 + 5 dx$$

$$du = 5(x^4 + 15 + 1) dx$$

$$\frac{du}{5} = \text{numerator } dx$$

$$\int \frac{du}{5} \cdot \frac{1}{u} = \int \frac{1}{5} \cdot \frac{1}{u} du$$

$$= \frac{1}{5} \int \frac{1}{u} du$$

$$= \frac{1}{5} \ln |x^5 + 25x^3 + 5x| + C$$

6. Determine if the integral is convergent or divergent. If convergent, evaluate it.

$$\int_5^{\infty} \frac{1}{\sqrt{x-4}} dx$$

$$u = x-4 \\ du = dx$$

$$\lim_{b \rightarrow \infty} \int_5^b (x-4)^{-\frac{1}{2}} dx$$

$$\int_a^b (u)^{\frac{p}{q}} du = \int u^{\frac{p}{q}} du = \frac{u^{\frac{p}{q}+1}}{\frac{p}{q}+1} = \frac{u^{\frac{p+q}{q}}}{\frac{p+q}{q}} \quad \begin{matrix} p = -1 \\ q = 2 \end{matrix}$$

plus C

$$\frac{2}{-3+2} u^{\frac{-3+2}{2}} + C = \frac{2}{-1} u^{-\frac{1}{2}} + C = -2 u^{-\frac{1}{2}} + C$$

$$= \frac{-2}{\sqrt{u}} + C = \frac{-2}{\sqrt{x-4}} \Big|_5^b$$

$$\frac{-2}{\sqrt{\infty-4}} - \frac{-2}{\sqrt{1}} = \frac{-2}{\infty} + 2$$

$$= 0 + 2 \\ = 2$$

7. Determine if the integral is convergent or divergent. If convergent, evaluate it.

$$\int_0^{\infty} \frac{x^2}{\sqrt{9+x^3}} dx$$

$$= \lim_{t \rightarrow \infty} \int_0^t \frac{x^2}{\sqrt{9+x^3}} dx$$

$$x^2 (9+x^3)^{-\frac{1}{2}} dx$$

$$u = 9+x^3$$

$$du = 3x^2 dx$$

$$\frac{1}{3} du = x^2 dx$$

$$\int (u)^{-\frac{1}{2}} \cdot \frac{1}{3} du$$

$$\frac{1}{3} \int u^{-\frac{1}{2}} du = \frac{1}{3} [2u^{\frac{1}{2}}]$$

$$\frac{d}{du} 2u^{\frac{1}{2}} = \frac{2}{2} u^{-\frac{1}{2}} = u^{-\frac{1}{2}}$$

$$= \frac{1}{3} \cdot 2 \cdot \sqrt{u} = \frac{2}{3} \sqrt{u}$$

$$= \frac{2}{3} \sqrt{9+x^3}$$

$$\lim_{t \rightarrow \infty} \left[\frac{2}{3} \sqrt{9+t^3} \right]$$

$$\lim_{t \rightarrow \infty} \left[\frac{2}{3} \sqrt{9+t^3} - \frac{2}{3} \sqrt{9+0} \right]$$

$$\lim_{t \rightarrow \infty} \left[\frac{2}{3} \sqrt{9+t^3} - \frac{2}{3} \sqrt{9} \right]$$

$$\lim_{t \rightarrow \infty} \left[\frac{2}{3} \sqrt{9+t^3} - \frac{2}{3} \cdot 3 \right]$$

$$\lim_{t \rightarrow \infty} \left[\frac{2}{3} \sqrt{9+t^3} - 2 \right]$$

$$\frac{2}{3} \sqrt{9+\infty^3} - 2$$

$$\frac{2}{3} \sqrt{\infty} - 2 = \infty$$

diverges

8

$$S1 \int_1^+ \frac{7}{x^3} dx$$

$$\int_1^+ 7x^{-3} dx = 7 \int_1^+ x^{-3} dx = 7 \left[-\frac{1}{2} x^{-2} \right]_1^+ = \left[-\frac{7}{2} x^{-2} \right]_1^+ = \left(-\frac{7}{2} +^{-2} \right) - \left(-\frac{7}{2} 1^2 \right)$$

$$= -\frac{7}{2} +^{-2} + \frac{7}{2}$$

$$S2 -\frac{7}{2} \left(\frac{1}{10} \right) + \frac{7}{2} \quad \text{where } t=10 \quad -\frac{7}{2} \left(\frac{1}{10^2} \right) + \frac{7}{2}$$

$$\left(-\frac{7}{2} \cdot \frac{1}{100} \right) + \frac{7}{2}$$

$$S3 -\frac{7}{2} \left(\frac{1}{1} \right) + \frac{7}{2} \quad \text{where } t=100 \quad \left(-\frac{7}{2} \cdot \frac{1}{100^2} \right) + \frac{7}{2}$$

$$S4 -\frac{7}{2} \left(\frac{1}{1} \right) + \frac{7}{2} \quad \text{where } t=1000 \quad \left(-\frac{7}{2} \cdot \frac{1}{1000^2} \right) + \frac{7}{2}$$

$$S5 \text{ The area under the curve is given by } \int_1^{\infty} \frac{7}{x^3} dx$$

$$= \lim_{t \rightarrow \infty} \int_1^t 7x^{-3} dx \quad 7 \cdot \frac{1}{x^2} = \frac{7}{x^2}$$

9 Determine if the int or cal is convergent or divergent:

$$\int_0^1 10 \frac{e^{\frac{1}{x}}}{x^3} dx$$

$e^{\frac{1}{x}}$ find smth smaller
 $e^{\frac{1}{x}} \geq 1$

$$\geq \int_0^1 10 e^{\frac{1}{x}} \frac{1}{x^3} dx$$

so you can see where this is problematic

Will be disc at 0 since undefined

cont on $(0, 1]$ and disc at 0

$$\lim_{t \rightarrow 0^+} \int_t^1 10 e^{\frac{1}{x}} \frac{1}{x^3} dx$$

$$= 10 \int_t^1 e^{\frac{1}{x}} x^{-3} dx$$

$$= 10 \int_t^1 \frac{1}{x^3} dx$$

$$= \int_t^1 x^{-3} dx$$

$$= \left[-\frac{1}{2} x^{-2} \right]_t^1 =$$

$$= -\frac{1}{2} \left[\frac{1}{x^2} \right]_t^1 = -\frac{1}{2} \left(\frac{1}{1^2} - \frac{1}{t^2} \right)$$

$\frac{1}{t^2} \rightarrow \infty$ as $t \rightarrow 0^+$

So at 0 this is ∞
 So it will be infinite

10. Determine if the integral is convergent or divergent using comparison theory

$$\int_0^{\pi} 14 \frac{\sin^2(x)}{\sqrt{x}} dx$$

satisfy divergence - $0 < u < w$

$$\int_0^{\pi} 14 \sin^2(x) dx$$

This is not actually bigger

using a function that is actually bigger

$$\frac{14 \sin^2(x)}{\sqrt{x}} \{0 < x < \pi\} < \frac{14}{\sqrt{x}} \{0 < x < \pi\}$$

$$\int_0^{\pi} \frac{14}{\sqrt{x}} dx = \lim_{t \rightarrow 0^+} \int_t^{\pi} \frac{14}{\sqrt{x}} dx$$

$$\lim_{t \rightarrow 0^+} 14 \int_t^{\pi} x^{-\frac{1}{2}} dx$$

$$\lim_{t \rightarrow 0^+} 14 \left[2x^{\frac{1}{2}} \right]_t^{\pi}$$

This will give answer

11 Find the values of p for which the integral converges
expressed as an inequality

$$\int_0^{\infty} \frac{43}{x(\ln(x))^p} dx$$

and evaluate the integral for those values of p

$$\int_0^{\infty} \frac{43}{x(\ln(x))^p} dx \text{ converges for } p > 1 \text{ and converges to } \frac{43}{p-1}$$

$$\int_e^{\infty} \frac{10}{x(\ln x)^p} dx$$

can be evaluated as the limit $\lim_{b \rightarrow \infty} \int_e^b \frac{10}{x(\ln x)^p} dx$
 which can be done with the substitution $u = \ln x$, for which $du =$

$$u = \ln x \quad \frac{du}{dx} = \frac{1}{x} \rightarrow du = \frac{1}{x} dx$$

With the substitution $u = \ln(x)$, the integration limit e ($x = e$) becomes $u = \ln(e) = 1$

Similarly, the integration limit ∞ for x becomes the limit ∞ for u

The substitution $u = \ln(x)$ makes the indefinite integral

$$\int_e^{\infty} \frac{10}{x(\ln(x))^p} dx = \int_1^{\infty} \frac{10}{(x(u))^p} dx$$

$$= \int_1^{\infty} \frac{10}{(e^u(u))^p} du$$

$$= \int_1^{\infty} \frac{10}{(e^u(u))^p} e^u du$$

$$= \int_1^{\infty} \frac{10 e^u}{(e^u(u))^p} du$$

$$= \int_1^{\infty} \frac{10}{(u)^p} du$$

$$\int_1^{\infty} \frac{10}{u^p} du \text{ converges to } \frac{10}{p-1} \text{ if } p > 1$$

$$u = \ln x \\ \ln x = u \\ x = e^u$$

$$du = \frac{1}{x} dx$$

$$\frac{du}{\frac{1}{x}} = dx$$

$$du \cdot x = dx$$

12 Determine if the integral is convergent or divergent. If convergent, evaluate it.

$$\int_2^{\infty} e^{-5p} dp$$

This one is simple enough to integrate

but w/ practice as $\int_2^{\infty} e^{-8p} dp$

$$v = -8p \quad \frac{dv}{dp} = -8 \quad dv = -8dp \quad -\frac{1}{8} dv = dp$$

$$\int_2^{\infty} e^v \left(-\frac{1}{8}\right) dv = -\frac{1}{8} \int_2^{\infty} e^v dv$$

$$= \lim_{b \rightarrow \infty} \int_2^b -\frac{1}{8} e^v dv$$

$$= \lim_{b \rightarrow \infty} -\frac{1}{8} \int_2^b e^v dv$$

$$= \lim_{b \rightarrow \infty} -\frac{1}{8} e^v \Big|_2^b$$

$$= \lim_{b \rightarrow \infty} -\frac{1}{8} e^{-8b} \Big|_2^b$$

$$= \lim_{b \rightarrow \infty} \left(-\frac{1}{8} e^{-8b} + \frac{1}{8} e^{-16} \right)$$

$$= \left(-\frac{1}{8} e^{-\infty} + \frac{1}{8} e^{-16} \right)$$

$$= 0 + \frac{1}{8} e^{-16} \quad \Rightarrow \text{basically } 0$$

$$\int_2^{\infty} e^{-5p} dp$$

$$v = -5p \quad -\frac{1}{5} dv = dp$$

$$\int_2^{\infty} e^v \left(-\frac{1}{5}\right) dv = -\frac{1}{5} \int_2^{\infty} e^v dv$$

$$\lim_{b \rightarrow \infty} -\frac{1}{5} \int_2^b e^v dv$$

$$= \lim_{b \rightarrow \infty} -\frac{1}{5} e^{-5p} \Big|_2^b$$

$$= \lim_{b \rightarrow \infty} \left(-\frac{1}{5} e^{-5b} + \frac{1}{5} e^{-10} \right)$$

$$= \left(-\frac{1}{5} e^{-\infty} + \frac{1}{5} e^{-10} \right)$$

$$= 0 + \frac{1}{5} e^{-10}$$